

Blind rigid-motion CBCT — the problem

one measurement, two unknowns: the geometry cannot be fitted without the image, and the image cannot be solved without the geometry

$$y = A_{P(\theta)}x + n, \quad P(\theta) = P_{nom}T(\theta)$$

the cone-beam forward model. A_P is the projector for the V per-view matrices P ; $T(\theta)$ is the rigid motion.

$$y \in \mathbb{R}^{V \times n_v \times n_u}, \quad x \in \mathbb{R}^N, \quad \theta \in \mathbb{R}^{V \times 6}$$

$V=360$ views, $n_v \times n_u = 500 \times 700$, $N=256^3$:
126 M measurements, 16.8 M + 2160 unknowns.

$$(\hat{x}, \hat{\theta}) = \arg \min_{x, \theta} \frac{1}{2} \|A_{P(\theta)}x - y\|^2 + \lambda \text{TV}(x) + \mathcal{R}_\psi(x)$$

the joint (blind) problem. It is BILINEAR in (x, θ) , hence non-convex: many minima explain the same y .

$$\Theta(x) \equiv \arg \min_{\theta} \frac{1}{2} \|A_{P(\theta)}x - y\|^2$$

GEOMETRY given the image. Measured here: from the cold FDK every estimator config plateaus at $\approx 2.0^\circ$; from a clean reference the same budget reaches $0.09\text{--}0.27^\circ$.

$$X(\theta) \equiv \arg \min_x \frac{1}{2} \|A_{P(\theta)}x - y\|^2 + \lambda \text{TV}(x)$$

IMAGE given the geometry. Measured here: the same loop reaches 40.45 dB with the true θ from step 0, 38.16 dB blind.

$$\hat{\theta} = \Theta(X(\hat{\theta})), \quad \hat{x} = X(\Theta(\hat{x}))$$

THE CIRCULARITY, stated exactly: each unknown is defined through the other, so neither map can be evaluated on its own.

$$A_{P(\theta \circ g^{-1})}(g \cdot x) = A_{P(\theta)}x \quad \forall g \in SE(3)$$

an EXACT gauge: a global rigid pose is unobservable, so the solution is an ORBIT $\{(g \cdot \hat{x}, \hat{\theta} \circ g^{-1})\}$, not a point.

$$\text{depth along the beam: } \partial y / \partial t_{\parallel} \approx 0$$

and beyond the gauge, translation ALONG THE BEAM is nearly unobservable — 92% of a converged estimator's residual lives there.

The scheme — a prior on the geometry bridge, then alternating minimization

the prior supplies the information the data cannot; the alternation is what makes each unknown computable from the other

Training (offline, on simulated motion)

$$x_t = \text{FDK}(y, P_{nom}T(t\theta)) + t\Delta, \quad \Delta = x_{static} - \text{FDK}(y, P_{nom}T(\theta))$$

the geometry bridge: $t=0$ is the uncorrected recon (the inference cold start), $t=1$ the motion-free scan.

$$\psi^\star = \arg \min_{\psi} \mathbb{E}_{t, \theta, x} \|v_\psi(x_t, t) - \dot{x}_t\|^2, \quad \dot{x}_t = \frac{d}{dt}x_t$$

flow matching against the EXACT tangent of the bridge, so v_ψ points along the set of reachable reconstructions.

Inference (blind; $k=0, \dots, N-1$, $t_k=k/N$, $\Delta t=1/N$, $N=50$)

$$\tilde{x}_k = x_k + \Delta t v_{\psi^\star}(x_k, t_k)$$

PREDICT — the prior moves first, evaluated on 64^3 tiles and blended back to the volume.

$$\theta_{k+1} \approx \arg \min_{\theta} L(A_{P(\theta)}\tilde{x}_k, y)$$

ESTIMATE — 50 warm-started Adam steps on a coordinate net over the view index. Note the argument: $\tilde{x}_k$, not x_k (Gauss–Seidel).

$$z_k = \arg \min_z \|A_{P(\theta_{k+1})}z - y\|^2$$

DATA — 5 CG iterations warm-started at $\tilde{x}_k$; the constraint set itself has just been re-aimed by (12).

$$x_{k+1} = z_k + \kappa(\text{TV}(z_k) - z_k)$$

PnP corrector, $\kappa=0.3$ — an OOD-safe denoiser, unlike using $v_{\psi^\star}$ itself inside the splitting.

$$\bar{\theta} = \frac{1}{K} \sum_{j=N-K+1}^N \theta_j, \quad x_{out} = \text{FDK}(y, P_{nom}T(\bar{\theta}))$$

readout, $K=2$: the estimator's tail is a period-2 limit cycle, so the mean is taken rather than the last step.